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Syllabus

MBA 532 - AI in Business

Instructor

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Course Description

Designed as an introduction to the evolving area of AI, this course emphasizes potential business applications and related managerial insights. Artificial Intelligence (AI) is the science behind systems that can program themselves to classify, predict, and offer solutions based on structured and unstructured data.

For millennia, humans have pondered the idea of building intelligent machines. Ever since, AI has had highs and lows, demonstrated successes and unfulfilled potential. Today, AI is empowering people and changing our world: Netflix recommends movies, Amazon recommends popular products, and several EV manufacturers are working to perfect self-driving cars that can navigate safely around other vehicles without human assistance. More recently, Generative AI (e.g., OpenAI's *GPT-4*, and variants of this concept such as Google's *Gemini*, Anthropic's *Claude* or Microsoft's *Copilot*) has revolutionized and energized imaginations and expectations with multi-modal capabilities. Businesses are scrambling to suitably adjust AI strategies across multiple domains and industries.

This course focuses on how AI systems understand, reason, learn and interact; learn from industry's experience on several AI use cases. It seeks to help students develop a deeper understanding of machine learning (ML) techniques and the algorithms that power those systems and propose solutions to real-world scenarios leveraging AI methodologies. Students will also learn the estimated size and scope of the AI market, its growth rate, and expected contribution to productivity metrics in business operations.

Course Outcomes

Upon successful completion of this course, you will be able to:

- Understand the history and evolution of AI, estimated AI market size, growth rate, impact on productivity metrics
- Understand the basics of how AI systems understand, reason, learn, and interact
- Learn through examples that highlight industry experiences through AI use cases
- Develop an understanding of machine learning (ML), Deep Learning, and other AI techniques/algorithms
- Explore the evolution of AI systems with multi-modal capabilities (vision, language, etc.) through technology convergence.
- Learn about AI state-of-the-art from media, industry, company, government, international agency reports
- Gain insights on real-world scenarios that leverage AI methodologies and technologies

Course Materials

The link to reading materials and resources to learn on the topics can be found in each week's learning module. All materials are available online for free, no required resources need to be purchased. There is no required textbook to supplement the course materials. Note: Be aware that some resources may open in a new tab.

Software Requirements: None

Course Outline

The course consists of 8 modules that focus on the following key areas:

Module 1: AI Overview/Landscape

Key concepts

- Lesson 1: What Does AI Represent or Mean? Reviewing Basic Definitions and AI history
- Students will be able to: understand AI's history; commonly used labels such as Narrow AI, Broad AI, Artificial General Intelligence (AGI); CPUs versus GPUs/NPUs versus IPUs, etc.; general purpose computing versus accelerated computing.
- Lesson 2: Size of AI Market and Growth Rate
    - Students will develop a comprehensive understanding of AI market size forecasts, AI growth rate, and other metrics.
- Lesson 3: Key Technology Catalysts for AI's Development
    - Students will understand the contributions to AI's Development story from computer chips (especially GPUs), the shift to accelerated computing, and the related transition from Retrieval-based computing to Retrieval augmented generative computing.

Readings

Lesson 1:

- What is AI? Can we distinguish between AI and non-AI systems?
- What is an NPU: the newAIchips explained
- The Age of AI has begun | Bill Gates
- 114 Milestones In The History OfArtificialIntelligence(AI)
- Accelerate AI/ML with the new IPU-based cloud platform
- An Insight into Accelerated Computing

Lesson 2:

- Generative AI to become $1.3 Trillion Market by 2032
- Artificial Intelligence (AI) Market
- AI leader says field's new territory is promising but risky
- 14 Tech Experts Predict Which Industries and Sectors will benefit the most from AI
- "Take Five - 5 Questions on AI

Lesson 3:

- Harnessing the Power of GPUs: A Game-Changer for Neural Networks
- Nvidia Hardware is Eating the World
- Two Computing Trends are emerging at once
- What is Retrieval Augmented Generation?

Module 2: Defining and Clarifying AI and Machine Learning Concepts

Key concepts

- Lesson 1: The Broad AI/Machine Learning Domain
    - Students will learn the broad components within the AI umbrella domain; understand the elements that comprise an AI system; learn about deployed AI systems in business, with examples
- Lesson 2: Understanding What Learning Truly Means in AI/ML
    - Students will understand how machine learning models differ from statistical models, their focus on prediction; types of learning in AI/ML (supervised, unsupervised, reinforcement, and deep learning with examples); basic elements and types of artificial neural networks; Convolutional Networks with examples
- Lesson 3:
    - Students will learn key factors in designing AI agents for AI/ML applications; System 1/System 2 thinking; Legal considerations in AI/ML; Aligning machine goals with human goals.

Readings

Lesson 1:

- What is AI?
- Machine Learning Explained
- AI vs. Machine Learning vs. Deep Learning vs. Neural Networks: What's the difference?
- Machine Learning Model Deployment
- How to Deploy Conversational AI
- How To Implement AI for Businesses

Lesson 2:

- Strong AI versus Weak AI: What is the Difference?
- Machine Learning versus Traditional Statistical Models
- What's the difference between supervised, unsupervised, semi-supervised, and reinforcement learning?
- What is the Turing Test?
- ChatGPT passed the Turing Test
- ANN vs CNN vs RNN

Lesson 3:

- System 1 and System 2 Thinking
- AI Agents: Exploring AI Agent Role, Design and Impact
- Aligning AI to Human Values Means Picking the Right Metrics

Module 3: AI and Technology Convergence

Key concepts

- Lesson 1: AI progress based on contributions from fully evolved technologies
    - Students will learn how fundamental, established/mature technologies that contributed to a growing set of AI capabilities.
- Lesson 2: AI progress based on contributions from midstream or still evolving, and emergent technologies
    - Students will learn how midstream (still evolving) technologies and emergent technologies that have increased AI capabilities.
    - Students will be able to: understand why AI has quickly acquired huge prominence and attention in recent years.
- Lesson 3: Kurzweil's Law of Accelerating Returns; Updated/new Turing Tests and Standards; other recent AI approaches such as affective computing and BCI (Brain-Computer Interface).
    - Students will learn about the latest AI contexts and how Kurzweil's Law, and new tests, standards and methods relate to technology convergence and AI progress.

Readings

Lesson 1:

- The Convergence of AI, NLP and Machine Learning Technologies: Driving Innovation in Language Technologies
- The Convergence of AI and Robotics
- The Convergence of AI and Computer Vision

Lesson 2:

- From Data to Decisions: The Transformative Power of Machine Learning
- The Convergence will be a powerful trend in tech
- The Convergence of AI and Other technologies
- OpenAI announces development of AI agents
- World's First AI-Designed CPU Paves Way For 'Self-Evolving Machines'

Lesson 3:

- AI chatbots learned to write before they could learn to think
- Audio Presentation by SoundHound at JMP Tech Conference March 4, 2024 (you be asked to register before accessing this presentation – please listen to the first 12 minutes of this presentation)
- AI-Powered 6G networks will reshape digital interactions
- Asimov's Three Laws of Robotics, Applied to AI
- What is the Law of Accelerating Returns? How It Leads to AGI
- SAE J3016 Levels of Driving

Module 4: AI abilities versus Human abilities, Human/Machine Collaboration

Key concepts

- Lesson 1: Assessing Human and AI abilities
    - Students will be able to: understand the history of human and AI abilities, including how to measure them, and why human abilities have remained relatively stable over time while machine abilities have advanced.
    - Students will be able to: learn about human-machine collaboration, and related concepts such as augmentation (leveraging human abilities and/or performance with AI/machines) versus automation (replacing humans with AI/machines).
- Lesson 2: Advantages and Disadvantages of AI
    - Students will be able to: recognize the advantages and disadvantages of AI correspond to the strengths and limitations of AI in comparison with human abilities and the five human senses.
- Lesson 3: AI/ML Use Cases
    - Students will be able to: learn a wide range of industry use cases that demonstrate the contribution of AI/ML to business operations.
    - Students will be able to: focus on approaches to break down the components of business decision-making such that the value added by machines and humans becomes clearly delineated.

Readings

Lesson 1:

- AI Capability List
- Strengthening the bonds of Human-Machine Collaboration, Deloitte report, November 2022
- Automation vs Augmentation: How Talent Acquisition Benefits from AI
- AI is no match for the quirks of Human Intelligence

Lesson 2:

- Siva K. Balasubramanian, "AI: The Land of Lingering Fear and Emergent Hope," Op-Ed commentary, TheWelNews, August 30, 2023
- What are the advantages and disadvantages of AI?
- Breaking down the advantages and disadvantages of AI
- Pros and Cons of AI (for the legal profession)

Lesson 3:

- 10 Everyday Machine Learning Use Cases
- Understanding AI/ML Use Cases
- 7 AI/ML Use Cases to Watch
- 12 Most Popular Use Cases in the Enterprise Today

Module 5: AI's impact on Work, Jobs, Humans, and Productivity

Key concepts

- Lesson 1: Generative AI-Driven Productivity Impact
    - Students will be able to recognize historical and widely used productivity metrics from the US Bureau of Labor Statistics, the Federal Reserve, and leading businesses (McKinsey).
    - Students will be able to recognize how Generative AI is expected to incrementally boost this historical metric and differences in Generative AI's impact across business areas/functions, countries, and wage quintiles are also assessed.
    - Students will be able to learn insights on Generative AI's productivity impact assessment from two widely cited research studies .
- Lesson 2: AI's impact on Job Losses
    - Students will understand that the positive impact from Generative AI on productivity metrics has to be interpreted in the context of negative impact in terms of significant job losses. Drawing on a variety of leading data sources (Goldman, Indeed, Stanford AI Index Report, and World Economic Forum), students will develop an understanding of Generative AI's differential impact across occupations and industries.
- Lesson 3: AI/ML models of human judgment, AI Chasm and three related gaps.
    - In part I, students will focus on the scope and challenges on efforts to model human judgment with machine learning, AI-based automation, and societal tradeoffs from AI/ML job impact; part II focuses on the AI Chasm and three related gaps (Gullibility gap, Illusory Progress gap, and Robustness gap.

Readings

Lesson 1:

- AI and the automation of work — Benedict Evans
- Future of work (to 2 online, (2017). Please click on the blue arrow to toggle between occupations to learn AI's impact on jobs by 2030 in six countries.
- Generative AI and the Future of work in America. Download the full report and read Chapter 2 (pages 22-29).
- Automation or augmentation? This is how AI will be integrated into the jobs of tomorrow.

Lesson 2:

- Everyday AI: Chatbots Are Rewriting How We Live and Work Today
- Choosing AI's Impact on the Future of Work
- AI in 2024: My top 5 Concerns and Top 5 Hopes

Lesson 3:

- AI Replacing Jobs statistics – The Impact on Employment in 2024
- Kochhar, R. (2023, July 26). Which U.S. Workers Are More Exposed to AI on Their Jobs?. *Pew Research Center's Social & Demographic Trends Project*
- 10 Top AI Jobs for 2024

Module 6: AI's Impact Assessment from other dimensions - Multiple Perspectives

Key concepts

- Lesson 1: Focus on friction points, worries, biases, challenges and limitations associated with AI
    - Part I will address friction points for businesses and AI, and Nations and AI. Part II will focus on mis-steps stemming about worrisome human predispositions (e.g., fundamental over-attribution error) and recent trends such as Sovereign AI; biases, challenges and limitations associated with AI/ML systems/algorithms.
- Lesson 2: Focus on AI strategy and deployment at top-ranked firms, and at specific business functions
    - Students will understand unique approaches toward AI deployment that are based on firm-specific strategies or innovations; this lesson also presents several examples and use cases specific to different business functions such as Retail, Sales, Accounting etc.
- Lesson 3: Focus on AI use cases in specific industries
    - Students will be presented with examples of AI use cases in Finance, Travel, Government, Education and Medical industries.

Readings

Lesson 1:

- How Countries can build an effective AI strategy
- What is Sovereign AI?
- Sources of bias in artificial intelligence that perpetuate healthcare disparities – A global review (please limit your reading to the following five sub-sections of this document: Background, Method, Results, Interpretation, and Author Summary)
- The 'AI Divide between the Global North and the Global South
- Countering the geographical impacts of automation: AI, and place disparities

Lesson 2:

- What Being an 'AI First' Company Means at Google
- How AI will change Retail
- What is artificial intelligence (AI) in finance?
- Accounting Efficiency with AI
- AI Hubs in the United States (Download and Read Full Data Brief)
- State-by-State AI Legislation Tracker

Lesson 3:

- 43 examples of AI in Education
- The Future Of eLearning: How AR, VR, And AI Are Changing The Game
- How AI is disrupting Medicine
- The potential for artificial intelligence in healthcare
- The State of AI in Different Countries – An Overview
- AI and Geopolitics

Module 7: Generative AI and Explainable AI

Key concepts

- Lesson 1: Definition and Narrative History of Generative AI
    - Students will be able to define Generative AI and learn the history of its evolution.
- Lesson 2: Progress, Growth and Benefits of Generative AI
    - Students will be exposed to growth forecasts specific to the Generative AI market, assess infrastructure support needed in training and inference for Generative AI, evaluate potential scope for innovations in Generative AI, and explore avenues for protecting privacy in Generative AI applications.
- Lesson 3: Explainable AI (XAI)
    - Students will be able to define what "Explainable AI" is and why it is important. They will be able to identify techniques and tools used to make AI decisions understandable to humans, and how XAI can be implemented in real-world AI systems to enhance transparency and accountability.

Readings

Lesson 1:

- Generative AI and the Future of Work, 2023 Deloitte report
- Explained: Generative AI
- Generative AI History
- Generative AI in 2024: The 6 most important consumer tech trends
- How Stitch Fix is using Generative AI to Help Us Dress Better

Lesson 2:

- Privacy Considerations for Generative AI
- 5 ways Generative AI Improves Data Privacy
- Ideas for Experimenting with Generative AI
- Spend on Generative AI will grow 36% annually to 2030

Lesson 3:

- What is Explainable AI?
- Explainable AI: The Key to Unlock Generative AI's potential
- Why Businesses Need Explainable AI
- Using Explainable AI in Decision Making Applications

Module 8: AI Ethics and Responsible AI

Key concepts

- Lesson 1: Overview of AI Ethics
    - Students will be able to: Define AI ethics, understand Anthropomorphism, tracking/surveillance by AI-based systems, and other moral/ethical dilemmas that arise in AI applications
- Lesson 2: Attention to Biases in AI
    - Students will learn the sweeping scope of AI-related biases that inadvertently creep into databases and algorithms that are widely used, importance of awareness, sensitivity and promptness in correcting biases.
- Lesson 3: Introduction to Responsible AI
    - Students will be able to define Responsible AI and its core principles such as fairness, transparency, accountability, and safety.
    - Students will be able to: apply these principles in the lifecycle of AI projects to ensure ethical and responsible AI development.
    - Students will be able to: apply steps recommended for AI practitioners by legal experts to manage Data, Privacy and Compliance; importance of developing a company-specific AI strategy.

Readings

Lesson 1:

- Anthropomorphic AI: An Overview
- How Anthropomorphism hinders AI education
- The Global Expansion of AI Surveillance
- AI Can Make Moral Judgments, but Should It? | Psychology Today

Lesson 2:

- AI Bias – What Is It and How To Avoid It
- Humans Absorb Bias from AI
- There's More to Bias Than Biased Data
- Which Ethical Implications of Generative AI Should Companies Focus On?
- The Ethics and Governance of Generative AI
- Many Executives Uncertain if Their Organizations Have Ethical Standards for Generative AI: Deloitte State of Ethics and Trust in Technology Report

Lesson 3:

- What is Responsible AI
- Responsible AI Practices
- AI Regulation in the U.S.: What's Coming, and What Companies Need to Do in 2023 | News & Insights | Alston & Bird
- Reported EU legislation to disclose AI training data could trigger copyright lawsuits
- American AI initiative annual report

Course Structure and Learning Activities

There are 8 content modules in this course and each module may take about 9-12 hours to complete. The final module consists of your final exam for the course.

This course is comprised of the following elements:

Learning Activities:

- **Readings**: Each module may include several required and/or supplemental readings.
- **Video Lessons**: In each module, the concepts you need to know will be presented through a collection of short videos. You may stream these videos for playback within the browser by clicking on their titles.
- **Practice Quizzes**: Each module will include some practice quizzes, intended for you to assess your understanding of the topics. You will be allowed unlimited attempts at each practice quiz. There is no time limit on how long you take to complete each attempt at the quiz. These quizzes do not contribute toward your final score in the class.

Assessments:

- **Discussion Forum**: This course has a place for you to interact with other learners about class-related topics.
- **Summative Module Assessments**: Each module will include at least one summative module assessment. You will be allowed one attempt every eight hours for each assessment. There is no time limit on how long you take to complete each attempt at the assessment. Your highest grade will be recorded.
- **Final Assessment**: This course will contain one summative course assessment that is organized into 2 parts. You will be allowed one attempt for each part. Before taking the exam, please make sure you are in a place with reliable internet connection. No retakes will be granted for the lack of internet access. You are in an online program and the use of the Internet is a requirement.

How to Pass This Course

Guidelines for completing and submitting each assigned course activity is posted along with the assignment. Assignments can be submitted at any time as you move through the module. No late assignments will be accepted. In case of extenuating circumstances beyond your control that prevent the submission of an assignment or exam, you have to enter a request with the program advisor and the instructor.

To qualify for a Course Certificate, simply start verifying your coursework at the beginning of the course and pay the fee. Coursera Financial Aid is available to offset the registration cost for learners with demonstrated economic needs. If you have questions about Course Certificates, please see the help topics here.

Also note that this course is part of the Master of Business Administration program offered by Illinois Institute of Technology. By earning a Course Certificate in this course, you are on your way toward earning a Specialization Certificate in this topic. See more information about the program here.

If you choose not to pay the fee, you can still audit the course. You will still be able to view all videos, submit practice quizzes, and view required assessments. Auditing does not include the option to submit required assessments. As such, you will not be able to earn a grade or a Course Certificate.

The following table explains the breakdown for what is required in order to pass the class and qualify for a Course Certificate. You must pass each and every required activity in order to pass this course.

Activity	Required?	Number per Course	Estimated Time per Module	% Required to Pass	% of Total Grade
Lecture Videos	Yes	3-6 per module	.5-1 hour	N/A	N/A
Practice Quizzes	No	3-6 per module	.5 hour	N/A	N/A
Discussions	No	1 per course	1 hour	N/A	N/A
Summative Module Assessments	Yes	1 per module	.5 hour	80%	7.5%/each module (60%)
Summative Course Assessments	Yes	1 -2 per course	1-3 hours	80%	40%

Letter Grades

Letter grades are used for the final grade. Information about IIT grading system can be found in the Graduate Student Handbook.

Letter Grade	Description	Points	Percentage
A	Excellent	4.00	90-100
B	Above Average	3.00	80-89.99
C	Average	2.00	70-79.99
E	Fail	0.00	Under 70%

Minimum Academic Standing

To maintain good academic standing, students must maintain a minimum cumulative GPA of 3.00. A minimum cumulative GPA of 3.00 is required to graduate from the MBA Program. If a student falls below a cumulative GPA of 3.00, the student is on Academic Probation and must work to bring their cumulative GPA back up to at least a 3.00 to graduate. If a student is not able to improve their GPA over time, they are at risk of academic dismissal.

Students must earn passing grades (A, B, or C) in the Core Requirements. If students earn a failing grade (E) in a Core Requirement, the student must re-take the course to earn a passing grade.

Getting and Giving Help

- Use the Learner Help Center to find information regarding specific technical problems. For example, technical problems would include error messages, difficulty submitting assignments, or problems with video playback. If you cannot find an answer in the documentation, you can also report your problem to the Coursera staff by clicking on the *Contact Us!* link available on each topic's page within the Learner Help Center.
- Use the flag icon under each item to report errors in lecture video content, assignment questions and answers, assignment grading, text and links on course pages, or the content of other course materials.
- Familiarize yourself with Coursera's policy on Accessibility.

Academic Integrity

Your attentiveness to academic integrity reflects the value you place on your own work and the work of others. In addition to Coursera's Honor Code, we also have high expectations for conduct during course participation.

Discussion Forums: Expectations

Sharing an online course with other avid learners like you gives you a unique opportunity to share, collaborate, and learn from others and their experiences, and helps you reinforce your understanding of the topics of the course. Interacting in the Discussion Forums is a great way to engage with your online community. We know that it is not possible to read every discussion forum post, so we recommend that you read those that interest you and reply when you can contribute. The forum is part of your class activities and everybody is expected to act professionally and be civil and respectful of others in your class. Failure to meet these expectations may be considered a break in the Academic Code of Conduct and may result in your removal from the course. Please, check tips and helpful tools to interact in discussion forums in this document.

Academic Code of Conduct

Above all else, learners are expected to ensure that their conduct helps to create an atmosphere conducive to learning and the interchange of knowledge. While it is understood that some of these items are subject to interpretation, learners should nonetheless endeavor to:

- Be respectful of fellow learners.
- Not discriminate against fellow learners in any manner.
- Conduct peer reviews in a timely manner and give useful feedback on what was done well, helpful suggestions for how to improve, and specific comments about why you gave the grade you chose to assist peers in their learning.
- Turn assignments in on time and follow instructions on all assignments including those affecting the use of technology.
- Be truthful in all communication, which includes, but is not limited to, avoiding academic dishonesty.

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SYLLABUS

Artificial Intelligence

Instructor

Course Description

Course Outcomes

Course Materials

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